FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Circular Laminae

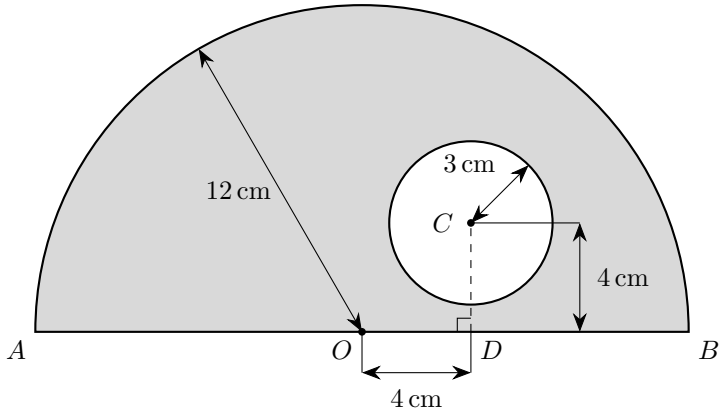
Worked solutions 7 questions · 76 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*9 marks*	2
Question 2	*9 marks*	4
Question 3	*8 marks*	6
Question 4	*9 marks*	8
Question 5	*12 marks*	10
Question 6	*14 marks*	12
Question 7	*15 marks*	15



1. A uniform semicircular lamina has radius 12 cm and diameter AB, with midpoint O.
A circular hole of radius 3 cm is cut from the lamina.
Its centre C is 4 cm above a point D on OB, where $OD = 4$ cm and CD is perpendicular to AB, as shown in the diagram above.

The remaining lamina is freely suspended from B and hangs in equilibrium.

Determine the acute angle that AB makes with the vertical, giving your answer to the nearest $0.1°$.

You may use the result that the centre of mass of a uniform semicircular lamina of radius r is on its axis of symmetry at a distance $\frac{4r}{3\pi}$ from its diameter. **[9]**

Solution

Take O as the origin, with the x-axis along OB and the y-axis perpendicular to AB, towards the semicircle. Measure coordinates in centimetres, so $B = (12, 0)$ and $C = (4, 4)$.

The centre of mass of a uniform semicircular lamina of radius r lies on its axis of symmetry, a distance $\frac{4r}{3\pi}$ from its diameter. Thus the centre of mass of the complete semicircle has coordinates

$$\left(0, \frac{4 \times 12}{3\pi}\right) = \left(0, \frac{16}{\pi}\right)$$

Let $\mathcal{A}_s$, $\mathcal{A}_h$ and $\mathcal{A}$ denote the areas of the complete semicircle, the hole and the remaining lamina respectively. Then

$$\mathcal{A}_s = \frac{1}{2}\pi(12)^2 = 72\pi \text{ cm}^2$$

$$\mathcal{A}_h = \pi(3)^2 = 9\pi \text{ cm}^2$$

$$\mathcal{A} = 72\pi - 9\pi = 63\pi \text{ cm}^2$$

Let the remaining lamina have centre of mass G, with coordinates $(\bar{x}, \bar{y})$. The lamina is uniform, so areas can be used as weights. If piece i has area $\mathcal{A}_i$ and centre of mass (x_i, y_i), then

$$\mathcal{A}\bar{x} = \sum \mathcal{A}_i x_i, \qquad \mathcal{A}\bar{y} = \sum \mathcal{A}_i y_i$$

where $\mathcal{A} = \sum \mathcal{A}_i$, with the area of a removed piece counted as negative.

Taking moments about the y-axis:

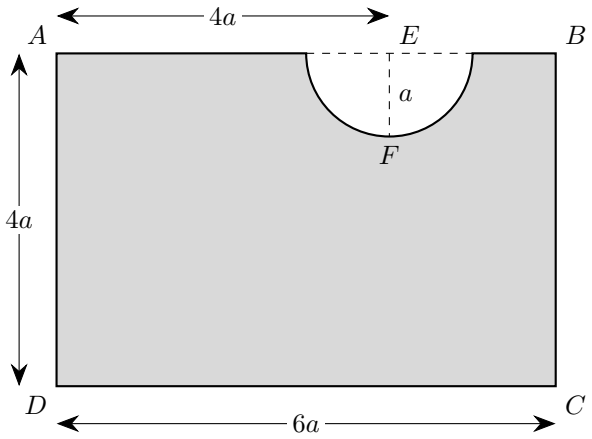
$$\begin{aligned} 63\pi\bar{x} &= 72\pi \times 0 - 9\pi \times 4 \\ &= -36\pi \\ \bar{x} &= -\frac{4}{7}\,\text{cm} \end{aligned}$$

Taking moments about the x-axis:

$$\begin{aligned} 63\pi\bar{y} &= 72\pi \times \frac{16}{\pi} - 9\pi \times 4 \\ &= 1152 - 36\pi \\ \bar{y} &= \frac{128 - 4\pi}{7\pi}\,\text{cm} \end{aligned}$$

When suspended in equilibrium, G lies vertically below B. Hence the required acute angle θ is the angle between BA and BG. Using the perpendicular distances from B to G:

$$\begin{aligned} \tan\theta &= \frac{\bar{y}}{12 - \bar{x}} \\ &= \frac{\dfrac{128 - 4\pi}{7\pi}}{12 + \dfrac{4}{7}} \\ &= \frac{32 - \pi}{22\pi} \\ \theta &= \arctan\left(\frac{32 - \pi}{22\pi}\right) \approx 22.7^\circ \qquad (1\text{ d.p.}) \end{aligned}$$



2. A uniform rectangular lamina $ABCD$ has $AB = 6a$ and $AD = 4a$, where $a > 0$.
A semicircular notch of radius a is cut from the edge AB.
The centre E of the semicircle lies on AB, with $AE = 4a$, and the semicircle lies inside the rectangle, as shown in the diagram above.

You may use the result that the centre of mass of a uniform semicircular lamina of radius r is a distance $\frac{4r}{3\pi}$ from its diameter.

(a) (i) Show that the perpendicular distance of the centre of mass of the remaining lamina from AD is

$$\frac{4(36-\pi)}{48-\pi}a$$

(ii) The remaining lamina is freely suspended from A and hangs in equilibrium. Calculate the acute angle that AB makes with the downward vertical, giving your answer to the nearest $0.1°$. **[9]**

Solution

(a) (i) Take D as the origin, with the x-axis along DC and the y-axis along DA. Let the centre of mass of the remaining lamina be $G(\bar{x}, \bar{y})$.

The area $\mathcal{A}$ of the remaining lamina is

$$\mathcal{A} = (6a)(4a) - \frac{1}{2}\pi a^2 = \frac{48-\pi}{2}a^2$$

The rectangle has its centre of mass at $x = 3a$. By symmetry, the removed semicircle has its centre of mass on the line through E perpendicular to AB, so its x-coordinate is $4a$.

The lamina is uniform, so areas can be used as weights. If piece i has area $\mathcal{A}_i$ and centre of mass (x_i, y_i), then

$$\mathcal{A}\bar{x} = \sum \mathcal{A}_i x_i, \qquad \mathcal{A}\bar{y} = \sum \mathcal{A}_i y_i$$

where $\mathcal{A} = \sum \mathcal{A}_i$, with the area of a removed piece counted as negative.

Taking moments about AD:

$$\mathcal{A}\bar{x} = 24a^2(3a) - \frac{\pi a^2}{2}(4a)$$

$$\frac{48-\pi}{2}a^2\bar{x} = (72-2\pi)a^3$$

$$\bar{x} = \frac{4(36-\pi)}{48-\pi}a$$

(ii) The centre of mass of a uniform semicircular lamina of radius r lies on its axis of symmetry, a distance $\frac{4r}{3\pi}$ from its diameter.

Thus the removed semicircle has its centre of mass at height $4a - \frac{4a}{3\pi}$ above DC. The original rectangle has its centre of mass at height $2a$. Taking moments about DC:

$$\mathcal{A}\bar{y} = 24a^2(2a) - \frac{\pi a^2}{2}\left(4a - \frac{4a}{3\pi}\right)$$

$$\frac{48-\pi}{2}a^2\bar{y} = 48a^3 - 2\pi a^3 + \frac{2}{3}a^3$$

$$\bar{y} = \frac{292 - 12\pi}{3(48-\pi)}a$$

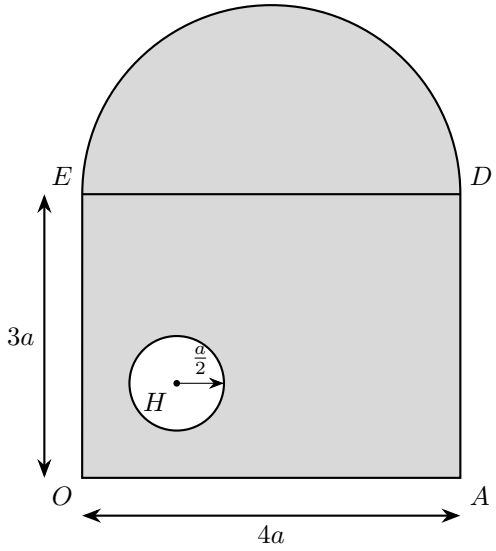
Let h be the perpendicular distance of G below AB. Then

$$\begin{aligned} h &= 4a - \bar{y} \\ &= \frac{12(48-\pi) - (292-12\pi)}{3(48-\pi)}a \\ &= \frac{284}{3(48-\pi)}a \end{aligned}$$

When the lamina hangs in equilibrium, G is vertically below A, so AG gives the downward vertical. Let θ be the acute angle between AB and AG. Using the right-angled triangle with components $\bar{x}$ parallel to AB and h perpendicular to AB, together with part (a)(i):

$$\begin{aligned} \tan\theta &= \frac{h}{\bar{x}} \\ &= \frac{284}{3(48-\pi)} \times \frac{48-\pi}{4(36-\pi)} \\ &= \frac{71}{3(36-\pi)} \end{aligned}$$

$$\theta = \arctan\left(\frac{71}{3(36-\pi)}\right) \approx 35.8^\circ \qquad (1 \text{ d.p.})$$



3. A uniform lamina is made by joining a rectangle $OADE$ to a semicircle along its diameter ED, and then removing a circular disc from the rectangular portion, as shown in the diagram above.

The rectangle has $OA = 4a$ and $OE = 3a$, where $a > 0$.
The removed disc has radius $\frac{a}{2}$ and centre H, which is a distance a from each of the sides OE and OA.

You may use the result that the centre of mass of a uniform semicircular lamina of radius r is a distance $\frac{4r}{3\pi}$ from its diameter.

(a) Find, in terms of a and π, the distance of the centre of mass of the remaining lamina from
(i) OE,
(ii) OA. **[6]**

The remaining lamina is freely suspended from A and hangs in equilibrium.
The angle between AO and the downward vertical is α.

(b) Find the exact value of $\tan\alpha$. **[2]**

Solution

(a) (i) Let the centre of mass be G, at distances $\bar{x}$ from OE and $\bar{y}$ from OA. The lamina is uniform, so areas can be used as weights. If piece i has area $\mathcal{A}_i$ and its centre of mass is at distances x_i from OE and y_i from OA, then

$$\mathcal{A}\bar{x} = \sum \mathcal{A}_i x_i, \qquad \mathcal{A}\bar{y} = \sum \mathcal{A}_i y_i$$

where $\mathcal{A} = \sum \mathcal{A}_i$, with the area of a removed piece counted as negative.

The semicircle has radius $2a$, so the remaining area $\mathcal{A}$ is

$$\mathcal{A} = (4a)(3a) + \frac{1}{2}\pi(2a)^2 - \pi\left(\frac{a}{2}\right)^2$$

$$= \frac{a^2(48+7\pi)}{4}$$

The rectangle and semicircle each have their centre of mass at distance $2a$ from OE; the removed disc has its centre at distance a. Taking moments about OE:

$$\begin{aligned}\mathcal{A}\bar{x} &= 12a^2(2a) + 2\pi a^2(2a) - \frac{\pi a^2}{4}(a) \\ &= \frac{a^3(96+15\pi)}{4} \\ \bar{x} &= \frac{a(96+15\pi)}{48+7\pi}\end{aligned}$$

(ii) The centre of mass of a uniform semicircular lamina of radius r lies on its axis of symmetry, a distance $\frac{4r}{3\pi}$ from its diameter. Here the diameter is $3a$ above OA, so the semicircle's centre of mass is at distance

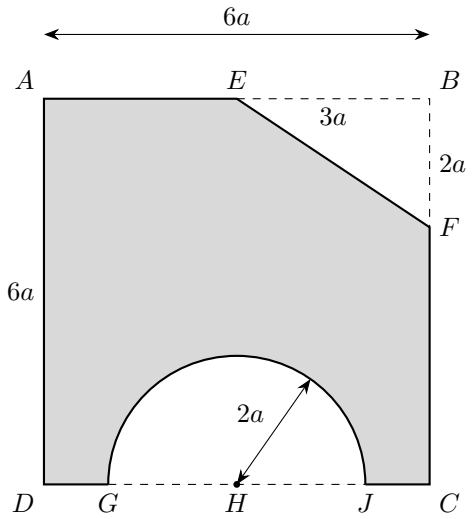
$$3a + \frac{4(2a)}{3\pi} = 3a + \frac{8a}{3\pi}$$

from OA. The rectangle's centre is at distance $\frac{3a}{2}$, and the removed disc's centre is at distance a. Taking moments about OA:

$$\begin{aligned}\mathcal{A}\bar{y} &= 12a^2\left(\frac{3a}{2}\right) + 2\pi a^2\left(3a + \frac{8a}{3\pi}\right) - \frac{\pi a^2}{4}(a) \\ &= a^3\left(18 + 6\pi + \frac{16}{3} - \frac{\pi}{4}\right) \\ &= \frac{a^3(280+69\pi)}{12} \\ \bar{y} &= \frac{a(280+69\pi)}{3(48+7\pi)}\end{aligned}$$

(b) In equilibrium, G lies vertically below A, so α is the angle between AO and AG. Relative to the original diagram, the distances from A to G parallel and perpendicular to AO are $4a - \bar{x}$ and $\bar{y}$, respectively. Then

$$\begin{aligned}4a - \bar{x} &= 4a - \frac{a(96+15\pi)}{48+7\pi} \\ &= \frac{a(96+13\pi)}{48+7\pi} \\ \tan\alpha &= \frac{\bar{y}}{4a-\bar{x}} \\ &= \frac{280+69\pi}{3(96+13\pi)}\end{aligned}$$



4. A uniform square lamina $ABCD$ has side $6a$.
A template T, shown shaded in the diagram above, is made by removing two portions of the lamina.

The first portion is the right-angled triangle EBF, where E lies on AB, F lies on BC, $EB = 3a$ and $BF = 2a$.

The second portion is a semicircle of radius $2a$.
Its centre H is the midpoint of DC, its diameter lies on DC, and the semicircle lies inside the square.

(a) Find the distances of the centre of mass of T from AD and from DC, giving both answers exactly in terms of a and π. **[6]**

The template is now freely suspended from A and hangs in equilibrium in a vertical plane.

(b) Find the acute angle between AD and the vertical, giving your answer to the nearest $0.1°$. **[3]**

Solution

(a) Take D as the origin, with the positive x-axis along DC and the positive y-axis along DA. Then $\bar{x}$ and $\bar{y}$ are the required distances from AD and DC, respectively.

The centre of mass of a uniform triangular lamina is $\frac{2}{3}$ of the way along a median from the vertex, so its coordinates are the means of the vertices' coordinates. Here $E(3a, 6a)$, $B(6a, 6a)$ and $F(6a, 4a)$, so the triangle's centre of mass is

$$\left(\frac{3a + 6a + 6a}{3}, \frac{6a + 6a + 4a}{3}\right) = \left(5a, \frac{16a}{3}\right)$$

The centre of mass of a uniform semicircular lamina of radius r lies on its axis of symmetry, a distance $\frac{4r}{3\pi}$ from its diameter. Thus the areas and centre-of-mass coordinates are

Portion	Area	x	y
Square	$36a^2$	$3a$	$3a$
Triangle removed	$3a^2$	$5a$	$\frac{16a}{3}$
Semicircle removed	$2\pi a^2$	$3a$	$\frac{8a}{3\pi}$

The lamina is uniform, so areas can be used as weights. If piece i has area $\mathcal{A}_i$ and centre of mass (x_i, y_i), then

$$\mathcal{A}\bar{x} = \sum \mathcal{A}_i x_i, \qquad \mathcal{A}\bar{y} = \sum \mathcal{A}_i y_i$$

where $\mathcal{A} = \sum \mathcal{A}_i$, with the area of a removed piece counted as negative.

The remaining area is

$$\mathcal{A} = 36a^2 - 3a^2 - 2\pi a^2 = (33 - 2\pi)a^2$$

Taking moments about AD and about DC:

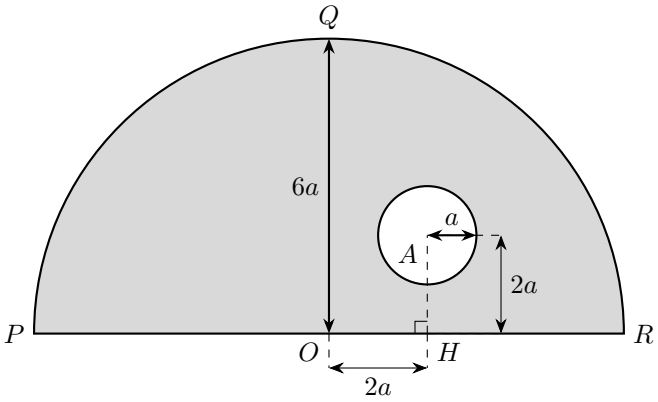
$$\begin{aligned}(33-2\pi)a^2\bar{x} &= 36a^2(3a) - 3a^2(5a) - 2\pi a^2(3a)\\ &= (93-6\pi)a^3\\ \bar{x} &= \frac{(93-6\pi)a}{33-2\pi}\end{aligned}$$

$$\begin{aligned}(33-2\pi)a^2\bar{y} &= 36a^2(3a) - 3a^2\left(\frac{16a}{3}\right) - 2\pi a^2\left(\frac{8a}{3\pi}\right)\\ &= \left(108 - 16 - \frac{16}{3}\right)a^3 = \frac{260a^3}{3}\\ \bar{y} &= \frac{260a}{3(33-2\pi)}\end{aligned}$$

(b) When the template hangs in equilibrium, its centre of mass is vertically below A. Let θ be the acute angle between AD and the vertical.

The centre of mass is $\bar{x}$ perpendicular to AD, and $6a - \bar{y}$ along AD from A. Using part (a):

$$\begin{aligned}\tan\theta &= \frac{\bar{x}}{6a-\bar{y}}\\ &= \frac{93-6\pi}{6(33-2\pi) - \dfrac{260}{3}}\\ &= \frac{279-18\pi}{334-36\pi}\\ \theta &= \arctan\left(\frac{279-18\pi}{334-36\pi}\right) \approx 45.2^\circ \qquad (1\text{ d.p.})\end{aligned}$$



5. A uniform semicircular lamina has centre O, radius $6a$ and diameter PR.
A circular disc of radius a is removed to form the lamina L shown shaded in the diagram above.
The centre of the removed disc is A.
The point H lies on OR, with $OH = 2a$, $HA = 2a$ and HA perpendicular to PR.

Take O as the origin, with the positive x-axis directed towards R and the positive y-axis directed towards Q.

(a) Find the exact coordinates of the centre of mass of L. **[6]**

The mass of L is M.
A particle of mass kM is attached at P.
The system is freely suspended from R and hangs in equilibrium, with P below R and the diameter PR making an acute angle of $\arctan\left(\frac{1}{6}\right)$ with the downward vertical through R.

(b) Find the exact value of k and the magnitude of the supporting force at R in terms of M and g. **[6]**

Solution

(a) The centre of mass of a uniform semicircular lamina of radius r lies on its axis of symmetry, a distance $\frac{4r}{3\pi}$ from its diameter.

Thus the original semicircle has area and centre of mass

$$\frac{1}{2}\pi(6a)^2 = 18\pi a^2, \qquad \left(0, \frac{4(6a)}{3\pi}\right) = \left(0, \frac{8a}{\pi}\right)$$

The removed disc has area πa^2 and centre of mass $A(2a, 2a)$. Therefore the area of L is

$$18\pi a^2 - \pi a^2 = 17\pi a^2$$

Let the centre of mass of L be $(\bar{x}, \bar{y})$. The lamina is uniform, so areas can be used as weights. If piece i has area $\mathcal{A}_i$ and centre of mass (x_i, y_i), then

$$\mathcal{A}\bar{x} = \sum \mathcal{A}_i x_i, \qquad \mathcal{A}\bar{y} = \sum \mathcal{A}_i y_i$$

where $\mathcal{A} = \sum \mathcal{A}_i$, with the area of a removed piece counted as negative.

Taking moments about the y-axis and about the x-axis:

$$17\pi a^2 \bar{x} = 18\pi a^2(0) - \pi a^2(2a) = -2\pi a^3$$

$$17\pi a^2 \bar{y} = 18\pi a^2\left(\frac{8a}{\pi}\right) - \pi a^2(2a) = (144 - 2\pi)a^3$$

Dividing by $17\pi a^2$, the exact coordinates are

$$\left(-\frac{2a}{17}, \frac{(144-2\pi)a}{17\pi}\right)$$

(b) Write $\theta = \arctan\left(\frac{1}{6}\right)$, so $\tan\theta = \frac{1}{6}$.

Relative to R, the centre of mass of L is a distance $6a - \bar{x}$ towards P and a distance $\bar{y}$ perpendicular to PR, on the Q side. Also, $RP = 12a$.

Resolve the weights parallel and perpendicular to PR. The parallel component of the lamina's weight has magnitude $Mg\cos\theta$, while the perpendicular components have magnitudes $Mg\sin\theta$ and $kMg\sin\theta$. Taking moments about R:

$$Mg\bar{y}\cos\theta = Mg(6a - \bar{x})\sin\theta + kMg(12a)\sin\theta$$

Dividing by $Mg\sin\theta$ gives

$$\bar{y}\cot\theta = 6a - \bar{x} + 12ak$$

Substituting the coordinates from part (a) and $\cot\theta = 6$:

$$6\left(\frac{(144-2\pi)a}{17\pi}\right) = 6a + \frac{2a}{17} + 12ak$$

$$\frac{(864-12\pi)a}{17\pi} = \frac{104a}{17} + 12ak$$

$$864 - 12\pi = 104\pi + 204\pi k$$

$$204\pi k = 864 - 116\pi$$

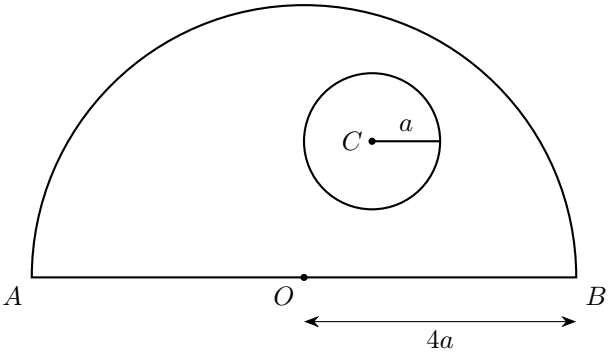
$$k = \frac{216 - 29\pi}{51\pi}$$

The supporting force F balances the total weight of the system, so

$$F = (M + kM)g$$

$$= \left(1 + \frac{216-29\pi}{51\pi}\right)Mg$$

$$= \frac{216 + 22\pi}{51\pi}Mg$$

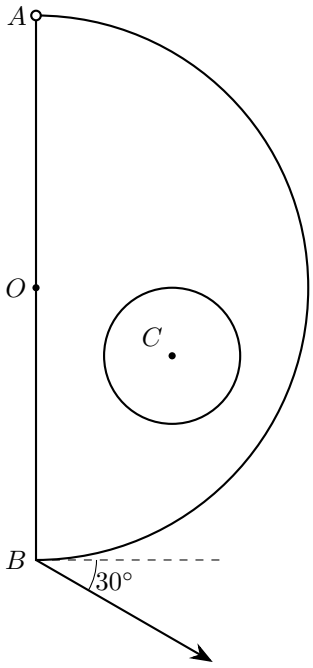


6. A uniform semicircular lamina has radius $4a$ and diameter AB, whose midpoint is O.
A circular hole of radius a is cut from the lamina.
In the orientation shown above, take O as the origin, the positive x-direction along OB and the positive y-direction perpendicular to AB towards the semicircular arc.
The centre C of the hole has coordinates $(a, 2a)$.

(a) Show that the centre of mass G of the remaining lamina has coordinates

$$\left(-\frac{a}{7},\ \frac{(128-6\pi)a}{21\pi}\right) \qquad \textbf{[6]}$$

The remaining lamina has mass 7 kg.
It is freely pivoted at A and turned so that AB is vertical, with A above B and the semicircular arc to the right of AB, as shown in the diagram below.
A light string attached to B holds the lamina in equilibrium.
The string lies in the plane of the lamina and makes an angle of 30° below the horizontal.



(b) Calculate the tension in the string. **[3]**

(c) Find the magnitude of the force exerted on the lamina by the pivot at A and the angle this force makes with the upward vertical, stating on which side of the vertical it acts. **[5]**

Solution

(a) The centre of mass of a uniform semicircular lamina of radius r lies on its axis of symmetry, a distance $\frac{4r}{3\pi}$ from its diameter. Thus the original lamina has centre of mass

$$\left(0, \frac{4(4a)}{3\pi}\right) = \left(0, \frac{16a}{3\pi}\right)$$

The areas of the original semicircle and the removed disc are

$$\frac{1}{2}\pi(4a)^2 = 8\pi a^2, \qquad \pi a^2$$

so the remaining lamina has area $7\pi a^2$.

Let the centre of mass of the remaining lamina be $G(\bar{x}, \bar{y})$. The lamina is uniform, so areas can be used as weights. If piece i has area $\mathcal{A}_i$ and centre of mass (x_i, y_i), then

$$\mathcal{A}\bar{x} = \sum \mathcal{A}_i x_i, \qquad \mathcal{A}\bar{y} = \sum \mathcal{A}_i y_i$$

where $\mathcal{A} = \sum \mathcal{A}_i$, with the area of a removed piece counted as negative.

Taking moments about the y-axis gives

$$7\pi a^2 \bar{x} = 8\pi a^2(0) - \pi a^2(a)$$

$$7\bar{x} = -a$$

$$\bar{x} = -\frac{a}{7}$$

Similarly, taking moments about the x-axis gives

$$7\pi a^2 \bar{y} = 8\pi a^2\left(\frac{16a}{3\pi}\right) - \pi a^2(2a)$$

$$7\bar{y} = \frac{128a}{3\pi} - 2a$$

$$\bar{y} = \frac{(128 - 6\pi)a}{21\pi}$$

Hence G has coordinates

$$\left(-\frac{a}{7}, \frac{(128 - 6\pi)a}{21\pi}\right)$$

(b) The lamina has been turned through $90°$ clockwise, so the original y-coordinate of G is now its horizontal distance to the right of AB.

The tension T has horizontal component $T\cos 30°$, acting at B, where $AB = 8a$. Its vertical component acts along AB, so has no moment about A.

Taking moments about A, and using part (a):

$$8aT\cos 30° = 7g\left(\frac{(128 - 6\pi)a}{21\pi}\right)$$

$$T = \frac{7g}{8\cos 30°} \times \frac{128 - 6\pi}{21\pi}$$

Using $g = 9.8\,\mathrm{m\,s^{-2}}$, the tension is

$$T \approx 16.4\,\mathrm{N}$$

(c) Let the pivot force have components R_x horizontally to the right and R_y vertically upwards. Resolving horizontally and vertically for equilibrium:

$$R_x + T\cos 30^\circ = 0$$

$$R_y - T\sin 30^\circ - 7g = 0$$

Thus the pivot force acts to the left and upwards, with components

$$R_x = -T\cos 30^\circ, \qquad R_y = 7g + T\sin 30^\circ$$

Using the unrounded value of T, its magnitude is

$$R = \sqrt{(T\cos 30^\circ)^2 + (7g + T\sin 30^\circ)^2}$$

$$\approx 78.1\,\text{N}$$

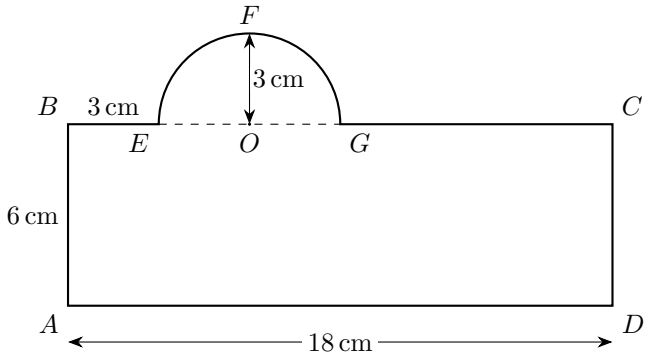
Let θ be the angle the force makes with the upward vertical. Then

$$\tan\theta = \frac{|R_x|}{R_y} = \frac{T\cos 30^\circ}{7g + T\sin 30^\circ}$$

$$\theta \approx 10.5^\circ \qquad \text{to the left of the upward vertical}$$

Final answers with $g = 10$: (b) 16.7 N; (c) magnitude 79.7 N

7. A uniform lamina consists of a rectangle $ABCD$ together with a semicircular tab EFG attached externally along its diameter EG, as shown in the diagram below. The two components have the same mass per unit area. The points E and G lie on BC, and O is the midpoint of EG.
The dimensions are $AB = 6\,\text{cm}$, $BC = 18\,\text{cm}$ and $BE = 3\,\text{cm}$. The semicircular tab has radius $3\,\text{cm}$.



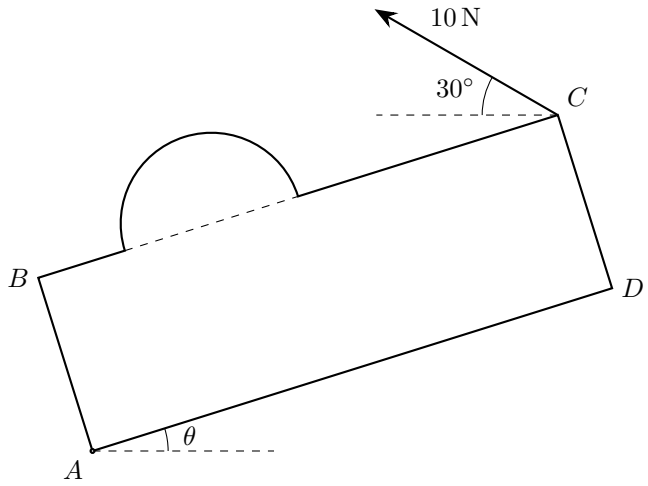
(a) Consider the semicircular tab EFG on its own. Using the standard centre-of-mass result for a uniform semicircular lamina, show that its centre of mass is $1.27\,\text{cm}$ from the diameter EG, correct to 3 significant figures. State the exact distance. **[2]**

(b) For the complete lamina, find

(i) the distance from AB to its centre of mass, giving your answer exactly, **[4]**

(ii) the distance from AD to its centre of mass, giving your answer exactly. **[3]**

(c) The complete lamina has weight $24\,\text{N}$ and is smoothly pivoted at A. A wire attached at C supports it in equilibrium. The wire pulls upwards and to the left at $30°$ above the horizontal, and its tension is $10\,\text{N}$.
The side AD makes an angle θ above the horizontal, where $0° < \theta < 45°$, as shown in the diagram below.



Calculate θ, giving your answer to 3 significant figures. **[6]**

Solution

(a) Consider only the semicircular tab EFG. The centre of mass of a uniform semicircular lamina of radius r lies on its axis of symmetry, a distance $\frac{4r}{3\pi}$ from its diameter. Here $r = 3\,\text{cm}$, so the distance from EG is

$$\frac{4 \times 3}{3\pi} = \frac{4}{\pi}\,\text{cm} \approx 1.27\,\text{cm} \qquad (3\text{ s.f.})$$

(b) The lamina is uniform, so areas can be used as weights. Let the centre of mass of the complete lamina be at distances $\bar{x}$ from AB and $\bar{y}$ from AD. If piece i has area $\mathcal{A}_i$ and its centre of mass is at distances x_i from AB and y_i from AD, then

$$\mathcal{A}\bar{x} = \sum \mathcal{A}_i x_i, \qquad \mathcal{A}\bar{y} = \sum \mathcal{A}_i y_i$$

where $\mathcal{A} = \sum \mathcal{A}_i$.

(i) The component areas and centre-of-mass distances are

Component	Area (cm^2)	Distance from AB (cm)
Rectangle	$18 \times 6 = 108$	$\frac{18}{2} = 9$
Semicircle	$\frac{1}{2}\pi(3)^2 = \frac{9\pi}{2}$	$BE + EO = 3 + 3 = 6$

Taking moments about AB:

$$\left(108 + \frac{9\pi}{2}\right)\bar{x} = 108 \times 9 + \frac{9\pi}{2} \times 6$$

$$\bar{x} = \frac{972 + 27\pi}{108 + \frac{9\pi}{2}}$$

$$= \frac{216 + 6\pi}{24 + \pi}\ \text{cm}$$

(ii) The rectangle's centre of mass is 3 cm from AD. Using part (a), the semicircle's centre of mass is at distance

$$\left(6 + \frac{4}{\pi}\right)\ \text{cm}$$

Taking moments about AD:

$$\left(108 + \frac{9\pi}{2}\right)\bar{y} = 108 \times 3 + \frac{9\pi}{2}\left(6 + \frac{4}{\pi}\right)$$

$$= 324 + 27\pi + 18$$

$$\bar{y} = \frac{342 + 27\pi}{108 + \frac{9\pi}{2}}$$

$$= \frac{76 + 6\pi}{24 + \pi}\ \text{cm}$$

(c) Resolve the tension parallel and perpendicular to AD. The angle between the wire and the direction DA is $\theta + 30^\circ$, so the components are

$$T_{\parallel} = 10\cos(\theta + 30^\circ)\,\text{N} \qquad \text{in the direction } DA$$

$$T_{\perp} = 10\sin(\theta + 30^\circ)\,\text{N} \qquad \text{in the direction } AB$$

These components both produce anticlockwise moments about A, with perpendicular distances 6 cm and 18 cm, respectively. The reaction at the pivot has zero moment about A.

After rotation through θ, the horizontal distance of the centre of mass from A is $\bar{x}\cos\theta - \bar{y}\sin\theta$. Taking moments about A, in N cm:

$$10[18\sin(\theta + 30^\circ) + 6\cos(\theta + 30^\circ)] = 24(\bar{x}\cos\theta - \bar{y}\sin\theta)$$

$$18\sin(\theta + 30^\circ) + 6\cos(\theta + 30^\circ) = \frac{12}{5}(\bar{x}\cos\theta - \bar{y}\sin\theta)$$

Expanding the compound angles and collecting terms:

$$18\left(\frac{\sqrt{3}}{2}\sin\theta+\frac{1}{2}\cos\theta\right)+6\left(\frac{\sqrt{3}}{2}\cos\theta-\frac{1}{2}\sin\theta\right)=\frac{12}{5}(\bar{x}\cos\theta-\bar{y}\sin\theta)$$

$$\left(9\sqrt{3}-3+\frac{12\bar{y}}{5}\right)\sin\theta=\left(\frac{12\bar{x}}{5}-9-3\sqrt{3}\right)\cos\theta$$

$$\tan\theta=\frac{\frac{12\bar{x}}{5}-9-3\sqrt{3}}{9\sqrt{3}-3+\frac{12\bar{y}}{5}}$$

Using the exact values from parts (b)(i) and (b)(ii), and $0^\circ<\theta<45^\circ$, gives

$$\theta\approx 17.4^\circ \qquad (3\text{ s.f.})$$